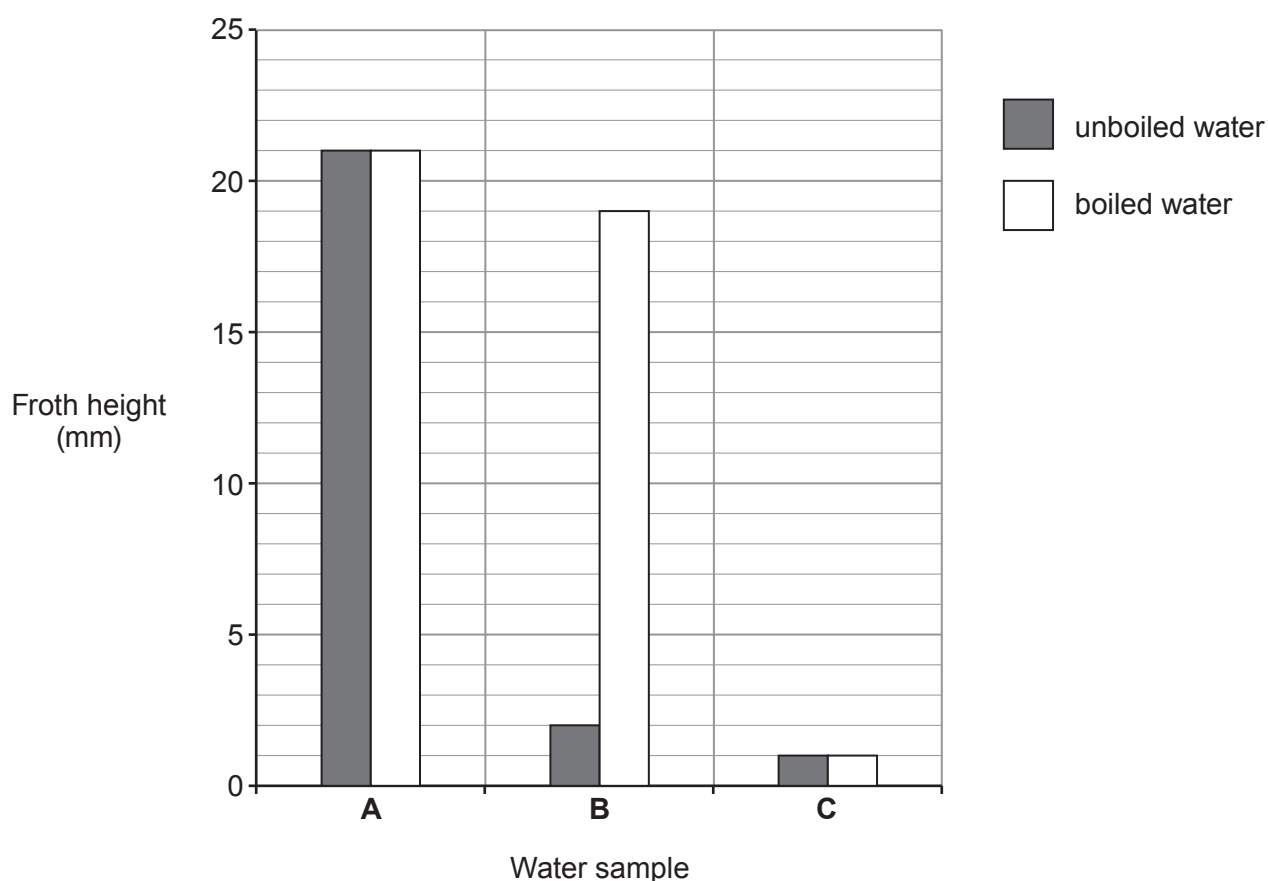


8. (a) Three samples of water, **A**, **B** and **C**, from different parts of the UK were tested in a laboratory.

1 cm³ of soap solution was added to 25 cm³ of the three different water samples. Each sample was shaken for 1 minute. The height of the froth was measured.

The experiment was repeated using new samples of water, **A**, **B** and **C**, that had been boiled.



Give the **letter** of the water sample which is

[2]

temporary hard water

permanent hard water

soft water



- (b) There are advantages and disadvantages of living in a hard water area.

Give **two** disadvantages of living in a hard water area.

[2]

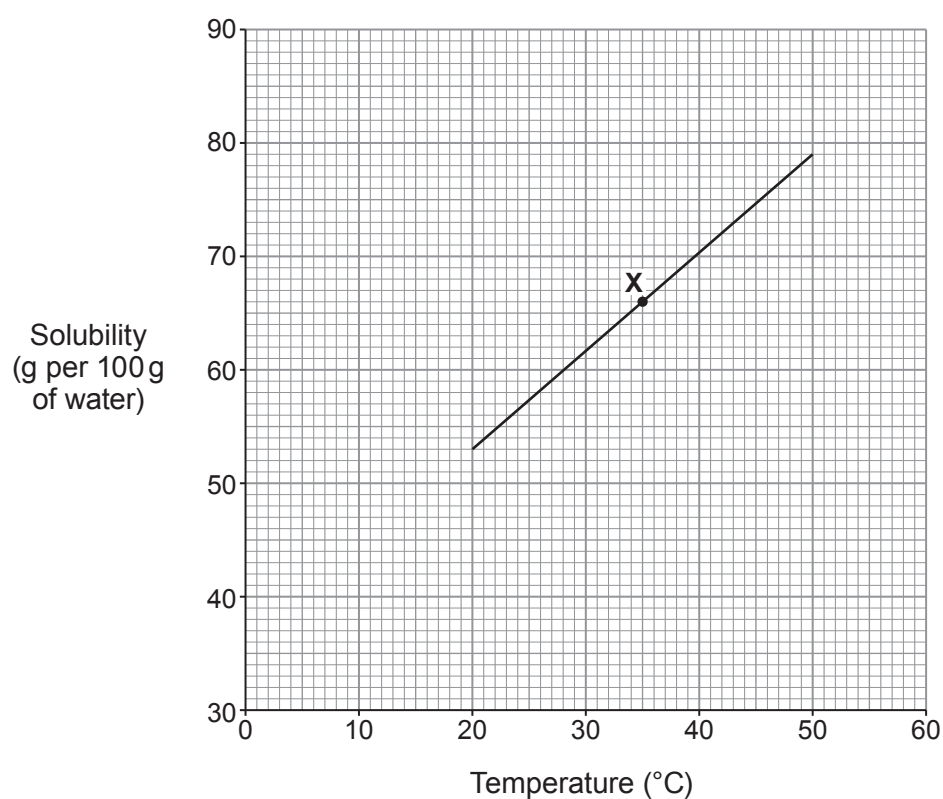
1

.....

2

.....

- (c) The graph below shows the solubility of lead nitrate in water at different temperatures.



- (i) State what point **X** on the graph tells you about lead nitrate.

[1]

.....

.....



- (ii) The solubility of lead nitrate at 20 °C is 53 g per 100 g of water.

Use the graph to find its solubility at 50 °C and hence calculate the mass of lead nitrate crystals that form when a saturated solution containing 100 g of water cools from 50 °C to 20 °C. [2]

Mass = g

- (iii) Use the graph to find the solubility of lead nitrate at 5 °C. [1]

..... g per 100g of water

